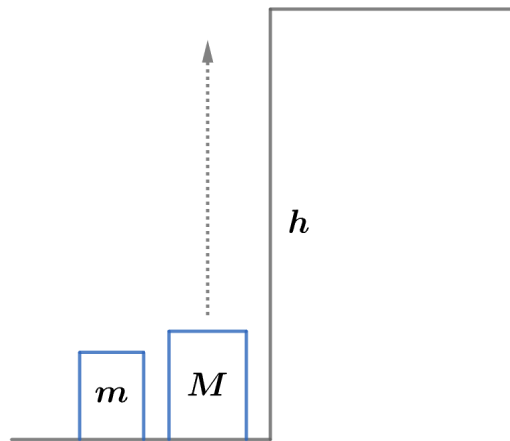
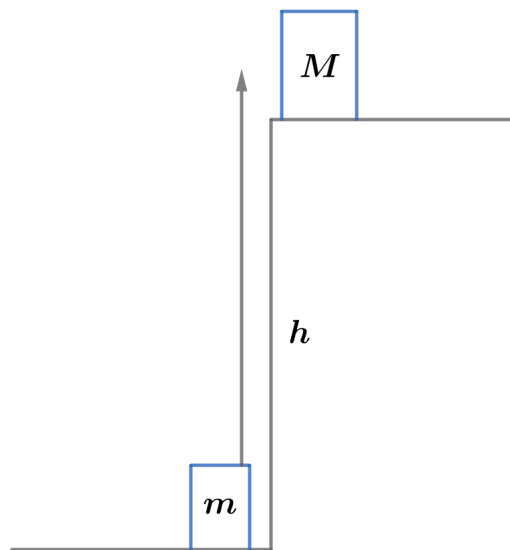


2009 F=ma Exam: Problem 12

Kevin S. Huang



Batman does work to increase his own gravitational potential energy by Mgh .



Then he does work to increase Robin's gravitational potential energy by mgh . The total work he does is

$$W = (M + m)gh = (100 \text{ kg} + 75 \text{ kg})(10 \text{ m/s}^2)(30 \text{ m}) = 5 \times 10^4 \text{ J}$$

so the answer is .